Report over a laboratory internship at McGill University

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1.1. Neutrinoless Double β-Decay in Xenon

Neutrinos are likely the most abundant massive particles of the Standard Model, yet they still hold secrets for modern physicist to discover. First postulated by Wolfgang Pauli[1], the "ghostly" particles have been a topic of research ever since.

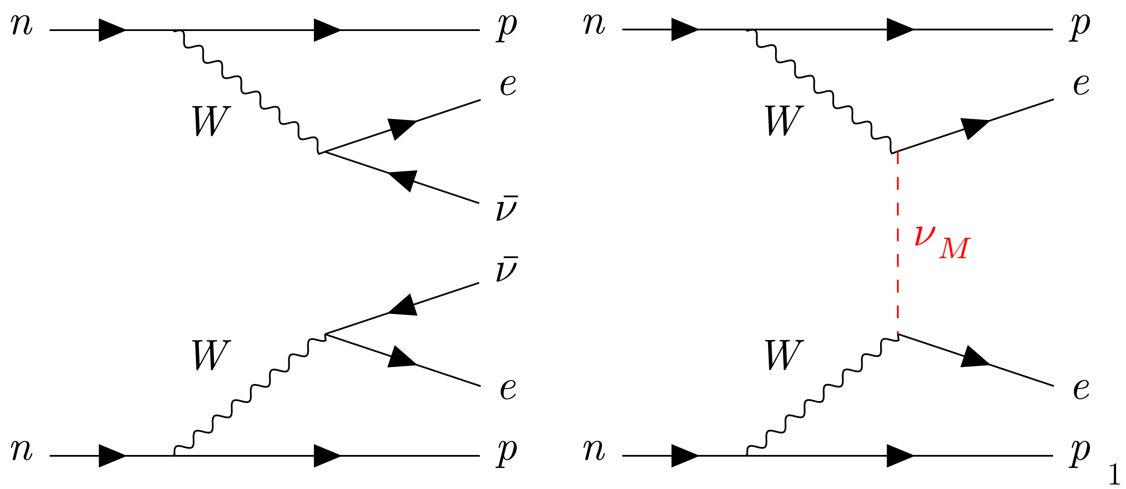
The experimental discovery of the neutrino by Clyde Cowan and Frederick Reines in 1956 [10] solved the problem of electron spectrum of the β decay and earned the two researches the Nobel Prize in Physics 1995. Likewise "the discovery of neutrino oscillations, which show that neutrinos have mass" was recognized with the Nobel Prize 2015.

Still, several challenges about the nature of neutrinos remain to this date. Despite the the establishment of neutrino masses, so far no experiment has been able to measure the absolute mass of either of the three neutrino flavour. Neutrino oscillation are only able to predict the squared mass difference between the three types of neutrinos, raising the additional question of the mass hierarchy (see fig. **??**).

Dirac versus Majorana Particles

One question connected to neutrino is particularly important for particle physicist because it puts the Standard Model of particle physics (SM) at stake. The SM assumes that all elementary fermions are clearly distinguishable from their anti-particles e.g. are

[1] "I have done a terrible thing, I have postulated a particle that cannot be detected." - Wolfgang Pauli in his letter to Gauvereinstagung in Tuebingen, 1930



(a) Two-Neutrino Double-β decay ($2\nu\beta\beta$) (b) Neutrinoless Double-β decay ($0\nu\beta\beta$)

Figure 1.1.: Quasi-Feynman diagrams for the Two-Neutrino Double-β decay (left) and the Neutrinoless Double-β decay (right). Shown is the lowest order process that just involves a virtual neutrino.

Dirac fermions. This property is trivial for charged particles, yet the absence of a electrical charge for neutrinos leaves the possibility of them being Majorana particles, particles that are identical with their anti-particle. Finding this to be the case would proof of physics beyond the SM as it shows that the Lepton number is in fact no conserved quantum number. It would also open the possibilty of neutrinos playing an important role in the matter-anitmatter asymmetry of the universe.

Neutrinoless double β-decay is a process that requires exactly this condition. Double beta decay corresponds to the simultaneous conversion of two neutrons to protons, a process that is in practise only measurable in even-even nuclides if single-β decay is forbidden due to energy conservation or at least highly spin-suppressed [7]. The ordinary double-β decay ($2\nu\beta\beta$) has so far been observed in thirteen isotopes and, including ^{136}Xe with the EXO-200 detector [2]. Figure 1.1 illustrates both ordinary and neutrinoless double β-decay.

Mechanism for $0\nu\beta\beta$

Among various possible mechanisms behind $0\nu\beta\beta$, the exchange of a light virtual Majorana neutrino ν_M (see figure 1.1b) is currently the most appealing due to two reasons: Firstly, experiments have shown the existence of three light massive neutrinos. Secondly, most theories assume the scale of new physics [implied by the observation of $0\nu\beta\beta$] to be much higher than the electroweak scale"[11].

The decay rate $\Gamma^{0\nu}$ of the $0\nu\beta\beta$ in the case of mediation by the exchange of a light virtual Majorana (electron) neutrino might be expressed as

$$\Gamma^{0\nu} = G^{0\nu}|M^{0\nu}|^2|\langle m_\nu\rangle|^2/m_e^2 \tag{1.1}$$

where $G^{0\nu}$ and $|M^{0\nu}|^2|$ are the phase space integral and the nuclear matrix element of the decay while $\langle m_\nu\rangle$ is the effective Majorana mass of the electron neutrino, calculated as the coherent linear of the the neutrino masses:

$$\langle m_\nu\rangle = \sum_{k=e,\mu,\tau} U_{e\,k}^2 m_{\nu_k} \tag{1.2}$$

where $U_{j\,k}$ are the elements of the Pontecorvo–Maki–Nakagawa–Sakata (PMNS) matrix and m_{ν_k} are the masses of the bare e, μ, and τ neutrino. If the observation of the $0\nu\beta\beta$ inside any material was possible, one could infer the absolute mass scale of the neutrinos.

It also needs to be stressed that even the non-observation of the $0\nu\beta\beta$ important constraints could be established: Assuming neutrinos to be Majorana particles, the inverted mass hierarchy could be ruled out by a negative result for $\langle m_\nu\rangle$ in the 20-30 meV range. Additionally, if the inverted mass hierarchy was to be demonstrated by other experiment, the same result would establish the the Dirac character of neutrinos.[11]

1.2. Overview over Experiments

While the existence of double β-decay was already hinted at through geochemical methods [15, page 43], $2\nu\beta\beta$ was first directly measured only in 1987 by the Irvine TPC in

^{82}Se [12]. Not soon after, the search for $0\nu\beta\beta$ began.

The general procedure is common for all experiments. The detector is loaded with a material that is known to double-β decay. In the case of $2\nu\beta\beta$ the energy spectrum for the electrons is continuous. However, in the absence or annihilation of the two neutrinos, the entire energy of the reaction will be converted into kinetic energy of the electrons. Therefore $0\nu\beta\beta$ will produce a peak at the Q-value of the reaction. Due to the long half-life of most isotopes against $0\nu\beta\beta$, generally several orders of magnitude larger than for $2\nu\beta\beta$, this peak is experimentally hard to observe and requires sophisticated background reduction techniques and highly sensitive detectors.

This section will give the reader a quick historical overview over past, current and proposed experiments.

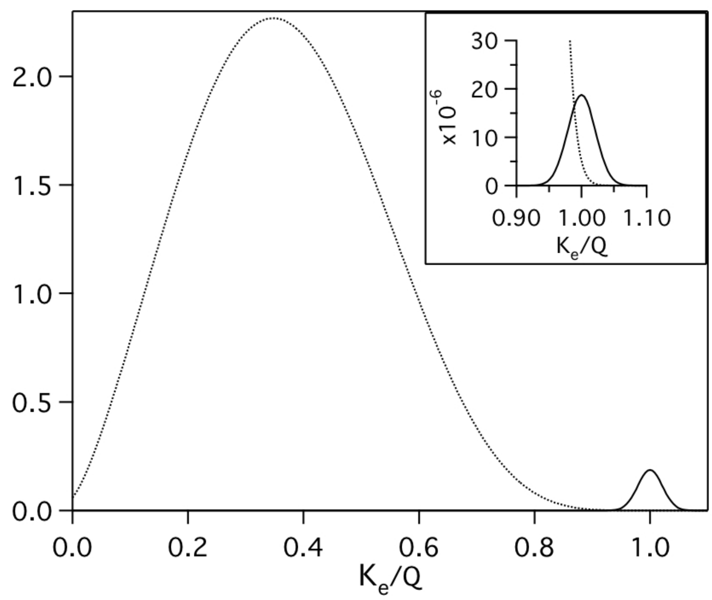


Figure 1.2.: Spectra of the sum of the electron kintetic energies K_e over the total Q-value of the reaction. The $2\nu\beta\beta$ spectrum (dotted line) is normalized to 1 while the $\nu\beta\beta$ spetrum (solid line) is normalized to 10^{-2} and 10^{-6} in the figure inset respectively. Both spectra assume a 5% energy resolution. From [13].

1.2.1. Non-Xenon Based Experiments

Cryongenic Underground Observatory for Rare Events (CUORE)

Building on the experience from the previous experiment Cuoricino and the prototype CUORE-0, CUORE will consist of 988 TeO_2 bolometers (total mass 741 kg, corresponding to 206 kg ^{130}Te). It will be placed inside a cryostate inside the Laboratori Nazionali del Gran Sasso, Italy (LNGS).

GERmanium Detector Array (GERDA)

Also situated in the LNGS, GERDA looks for $0\nu\beta\beta$ in ^{76}Ge. The experiment is currently in phase-II. The active mass of 35.8 kg of high purity Ge crystal diodes serves as both decay source and particle detector. Long term plans for the third phase, a tonne-scale experiment, that could push the sensitivity below the inverted hierarchy mass threshold.

Other experiments, such as the MAJORANA Demonstrator also use ^{76}Ge, yet take a different approach. For a tonne scale Germaninum experiment efforts from various collaboration likely have to be joined. Therefore most experiments also serve the purpose of finding the best candidate technology.

SNO+

SNO+ uses the structure and existing equipment that famously established neutrino-oscillations at SNOLAB, Sudbury, Canada. The heavy water inside the acrylic vessel has been replaced 780 tonnes liquid scintillator that will be loaded with neodymium salt (corresponding to 50kg ^{150}Nd). ^{150}Nd has the highest expected $0\nu\beta\beta$ decay rate due to its large phase scale factor. [15, page 51]

1.2.2. Xenon-based Experiments

The search for $0\nu\beta\beta$ in ^{136}Xe can be subdivided into three categories:

(a) Enriched ^{136}Xe dissolved in liquid scintillator

(b) High-pressure gaseous Xenon

(c) Liquid Xenon

KamLAND Zen

Similar to SNO+, the existing KamLAND detector at Kamioka Observatory, Toyama, Japan that has provided the currently best parameters for neutrino oscillation has been modified to enable search for $0\nu\beta\beta$. A extremely radiopure and transparent balloon containing enriched ^{136}Xe (phase I: 320 kg; phase II: 380 kg) dissolved in a decan bases liquid scintillator is placed inside 1000 ton liquid scintillator detector [23].

Neutrino Experiment with a Xenon TPC (NEXT)

The NEXT collaboration works towards a 100 kg gaseous ^{136}Xe detector (NEXT-100), to be operated inside the Laboratorio Subterraneo de Canfranc (LSC), Spain. Two prototypes of the high pressure gaseous xenon TPC were build to demonstrate the high energy resolution ($\lesssim 1\%$ at FWHM) of the NEXT design. Currently the collaboration is operating NEXT-NEW, a scale 1:2 in size (1:8 in mass) prototype of NEXT-100 build with the same radiopurity requirements as the proposed detector at LSC.

1.3. Enriched Xenon Observatories (EXO)

The work presented in this report was conducted in the laboratory of Thomas Brunner at McGill University, Montreal, Canada. He is part of the EXO collaboration. Therefore a particular focus will be put on the two experiments EXO-200 (current) and nEXO (proposed).

1.3.1. EXO-200

EXO-200 is the current experiment of the collaboration, located at the Waste Isolation Plant Project (WIPP) in New Mexico, USA. The detector shown in figure **??** is time-projection chamber (TPC) loaded with 175 kg of enriched liquid xenon ($\sim 81\%$ ^{136}Xe).

The TPC is divided by a optically transparent, common cathode, creating two almost identical drift regions (electric field $\sim$ 400V/m) [8]. Large-area avalanche photo diodes (APDs) and two wire planed (u and v wires) behind the anodes on either end of the detector read both the photons and the electric charge. Events consist of radioactive decays and cosmic radiation that deposit energy inside detector volume, resulting in the ionization of xenon atoms and the creation of scintillation light and free electrons that are drifted towards the anode wire planes.

Simultaneous measurement of scintillation light and charge allows full reconstruction of the energy, the location within the detector volume, and the multiplicity[2] for each event. While β events are mostly single site events (energy deposit at only one position), scattered γs are multi-site events (energy deposit at multiple locations). α events are identified through events with mostly scintillation light and little to none charge. Determining the position helps to optimize the sensitivity of the detector as the fiducial volume can be adjusted and one self-shielding of xenon can be taken advantage of.

During the two-year long phase I of the experiment, EXO-200 achieved an energy resolution of $1.53 \pm 0.06\%$ at $Q_{\beta\beta}$ [4]. This data collected during phase I allowed to directly measure the half life of ^{136}Xe against $2\nu\beta\beta$, $T^{2\nu}_{1/2} = 2.165 \pm 0.061 \times 10^{21}$ years[3] [3] while setting a lower limit for the $0\nu\beta\beta$ half life $T^{0\nu}_{1/2} < 1.1 \times 10^{25}$ years at 90 % confidence level (C.L.).

1.3.2. nEXO (next EXO)

Based on the success of EXO-200 the collaboration has proposed a tonne-scale experiment, nEXO that will likely be located in a cryopit at SNOLAB. Please see figure 1.3 for an artist rendering of the detector and its deployment inside the SNOLAB cryopit.

Instead of having the cathode placed in the middle of the TPC like in EXO-200, the nEXO design envisions a cylindrical, monolythic, single volume, liquid xenon TPC. The cathode and anode on either end will create a drift field of $\sim 400\,\mathrm{V/cm}$. The TPC will contain around 5 tonnes of enriched xenon ($\sim 90\%$ of ^{136}Xe).

The charge signal will be collected by a segmented anode while the scintillation light will be recorded using Silicon Photo Multipliers (SiPMs). They will be mounted behind

[2] Number of locations at which energy was deposited.
[3] This is the slowest decay rate, ever to be measured directly by any experiment.

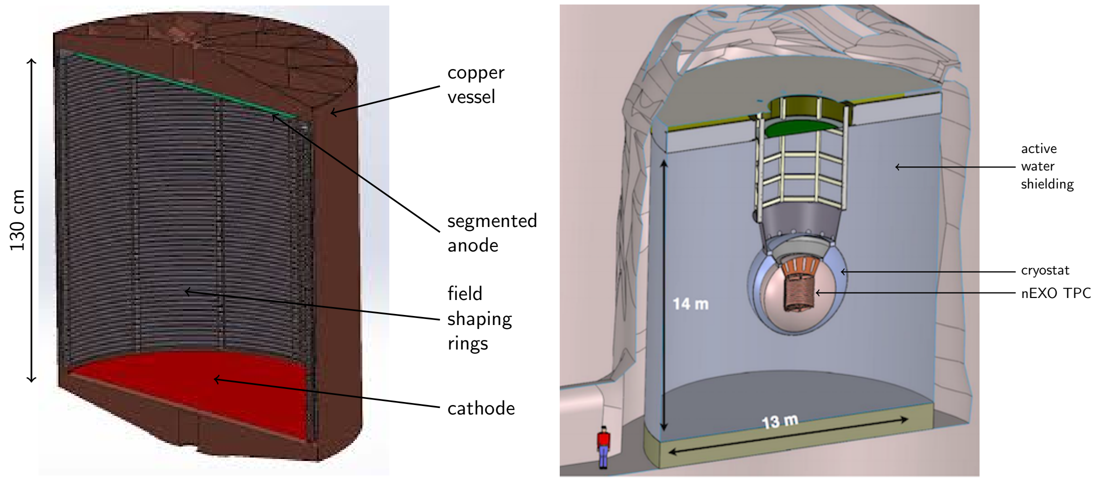


Figure 1.3.: Artist rendering of the nEXO TPC (left)[28] and its installation in the SNOLAB cryopit (right)[18].

the field shaping rings (see fig. 1.3a), but still inside the LXe volume, covering the entire cylinder surface (area $\sim 4\,\mathrm{m}^2$).

The large surface area made it necessary to find an alternative for the APDs, as they cannot be obtained in sufficient quantity. Highest radiopurity requirements forbid the use of PMTs. The collaboration currently invests many resources on the development and testing of SiPMs devices sensitive to LXe scintillation light at 175 nm. The centre part of my internship at McGill University was the design of a light source that would serve as a test setup for SiPMs.

Based on EXO-200 and radio-assay data, nEXO has been simulated and its sensitivity to $T^{0\nu}_{1/2}$ after 10 years of data acquisition has been predicted to be 9.5×10^{27} years. The simulations assumed that better light detection and electronics will help to improve the energy resolution of the detector to about 1% at $Q_{\beta\beta}$. Like its predecessor EXO-200, nEXO will deploy a full reconstruction of the energy, location multiplicity and topology of each event. This combined with the self-shielding of xenon enables nEXO to achieve a significantly higher sensitivity than other experiments using similar target masses of a different isotope.

1.3.3. Current and Projected Results

Using the current half-life limits of EXO-200 and KamLAN-Zen combined with the nuclear matrix elements computed in [19, 21], it is possible to define the allowed region for the effective Majorana neutrino mass for both the normal and inverted mass hierarchy. Figure 1.4 shows this permitted parameter space as function of the lightest neutrino mass eigenstate m_{min}. Furthermore, the figure incorporates the currently measured sensitivities for EXO-200 (after phase-I [4] and phase-II [5]) as well as the projected sensitivity for nEXO.

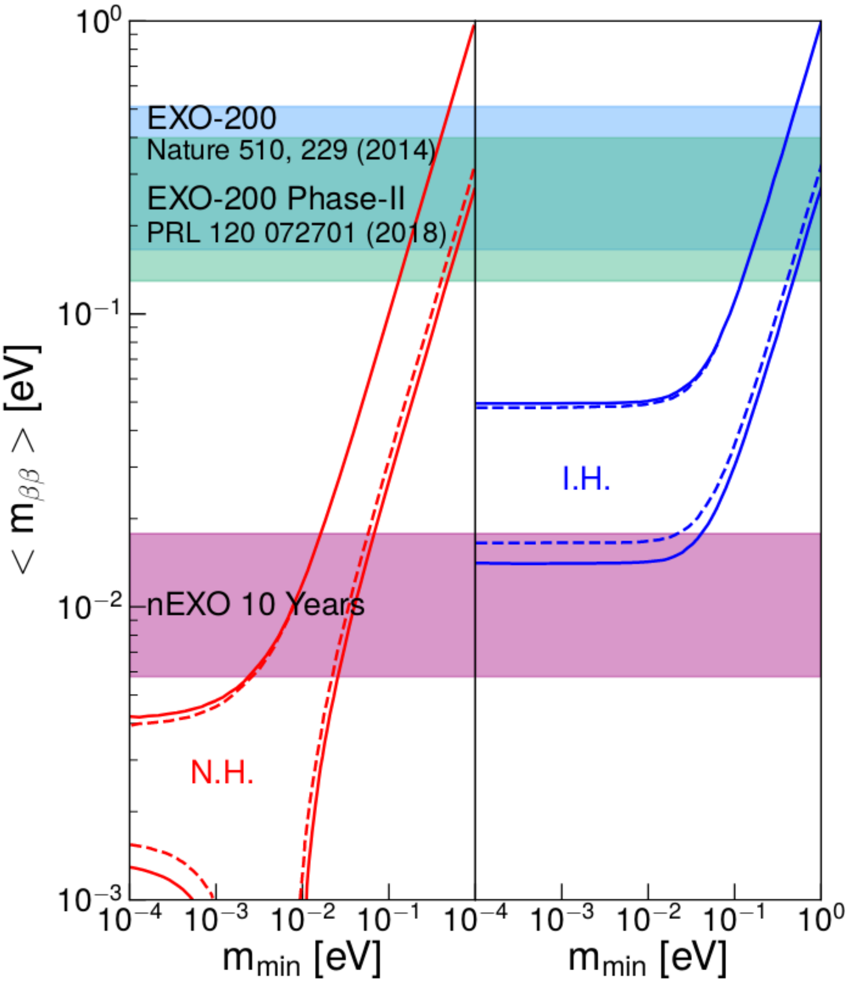


Figure 1.4.: 90% C.L. effective Majorana neutrino mass $m_{\beta\beta}$ exclusion sensitivity as function of the lightest neutrino mass m_{min}. Blue band: EXO-200 phase-I; green band: EXO-200 phase-II; purple band: nEXO (projected).
Uncertainties in the nuclear matrix elements lead to the width of the bands under the assumption $g_A = 1.27$. The inner dashed lines result from the unknown, irreducible Majorana phases. The outer solid lines incorporate the 90% C.L. errors of the three-flavor neutrino fit. Left: normal hierarchy; Right: inverted hierarchy. Image from [6].

2. Data structure

The three readout channels of the light source produce a analog signal that will be processed using the CAEN DT5730 digitizer. For storing the acquired data a set of object classed was created.
The classes `Event`, `Channel`, `TriggerInfo`, and `RawBlock` are all `TObject` classes that can be used within the `ROOT` framework. The following chapter will lay out how they are utilized for the data acquisition.

2.1. Class: **`RawBlock`**

When recording data with the digitizer, the contents of the acquisition window or parts of it might be written into a **`RawBlock`**. This primitive data structure consist of a vector containing the samples read from the digitizer and the bin number (nBin) of the first sample.

When recording full waveforms nBin is always zero. Yet, in the case of ZLE suppression, nBin contains the crucial information where to locate the following samples within the acquisition window.

2.2. Class: **`Channel`**

`Channel` object summarizes the data readout from one channel of the digitizer during one acquisition cycle. It contains the channel number and a list of the **`RawBlock`**s. If ZLE suppression is not in use this list simply has only one entry.

2.3. Class: **`Event`**

Finally, the multiple **`Channel`** objects acquired during one readout cycle are combined into one **`Event`** object that then is stored in a **`TTree`**. For the identification each **`Event`** object within a run is assigned a unique identification number (evtID). Furthermore, additional information about the Trigger for the event are stored in a **`TriggerInfo`** object.

3. Design of electroluminescent light source using a Cf-252 fission source

3.1. Objective

Testing and characterizing SiPMs for the use in nEXO is challenging for several reasons. One of the most dominant is the relative absence of reliable light sources that produce UV light at 175-178 nm. The light source designed during this summer internship is one of two different designs currently developed by the nEXO group at McGill. Both designs produce light in this region through scintillation in gaseous Xenon.

The decision to use a Cf-252 fission source for the light production was made prior to my arrival in Montreal having especially measurements and analysis of the SiPM efficiency in mind. On one hand, by using a calibrated fission source, one has good knowledge of the activity of the source and thus the total expected event rate. On the other hand, the fission source has the capability to produce a signal in multiple readout port, making a direct measurement of the relative efficiency of a SiPM in comparison to better understood detectors like PMT tubes possible.

The second light source takes a different approach. The scintillation light will be produced by electrons created from a photo-cathode after irradiation by a UV LED. Primarily responsible for this source was another student in the group, Antoine Belley whose report can be found here:

3.2. Working principle

Californium is a radioactive, synthetic element that can be produced via the irradiation of other actinides. First discovered in the Lawrence Berkley National Laboratory at the University of California in 1950 [25], Currently only two facilities worldwide have sufficient neutron flux to produce Cf via the the irradiation of Curium (Cm), the Oak Rigde National Laboratory and the Research Institute of Atomic Reactors in Dimitrovgrad, Russia [14].

^{252}Cf is the isotope used for the ESL. It has a average half life of 2.645 years. It's primary decay mode are α-decay and spontaneous fission with branching ratios of 96.91%, and 3.09% respectively [1].

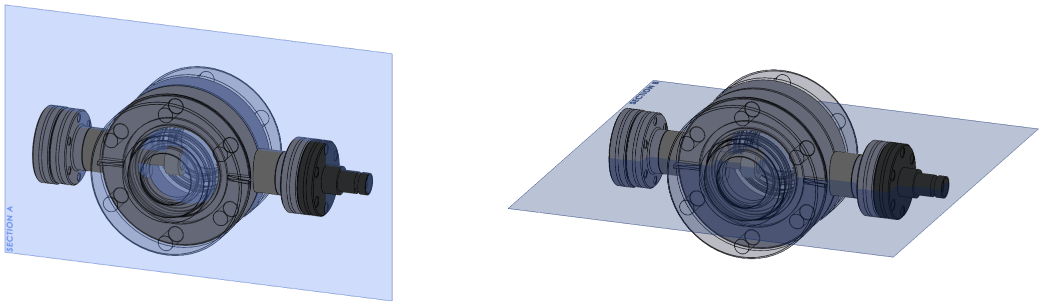
3.3. Design

Using the 3D-CAD software *Solidworks* a model of the lamp was design that served several purposes. I was building upon preliminary designs made prior to my arrival in Montreal. The lamp should feature at least two ports for light output (e.g. windows) and one additional port for a charge readout that could be used for coincident measurements and to discriminate against signals not arising from the scintillation light produced inside the lamp.

Figure 3.1 is a 3D rendering of the final design of the lamp while figure 3.2a and 3.2b shows the inside of the lamp to scale. Those drawings where also used for the the customization of parts carried out by the machine shop at *Université de Montréal*[1]

Certain issues had to be paid special attention to during the design that will be touched upon in the following.

[1]Université de Montréal, Laboratoire René-J.A.-Lévesque, 2905 Chemin de Service, Montréal, QC H3T 1J4



(a) Section A (b) Section B

Figure 3.1.: Rendering of the lamp including the two cuts for the section views depicted in figure 3.2.

3.3.1. Vacuum compatibility

Vacuum compatibility was important for the design of the lamp for two reasons. Firstly, the lamp should, once filled with Xenon, produce light at a steady and predictable level without refilling. This does not allow for leakage out of the lamp, as changing concentration of Xenon would greatly distort the light output that else only depends on the activity of the fission source. Using vacuum flanges and other parts was thus obligatory.

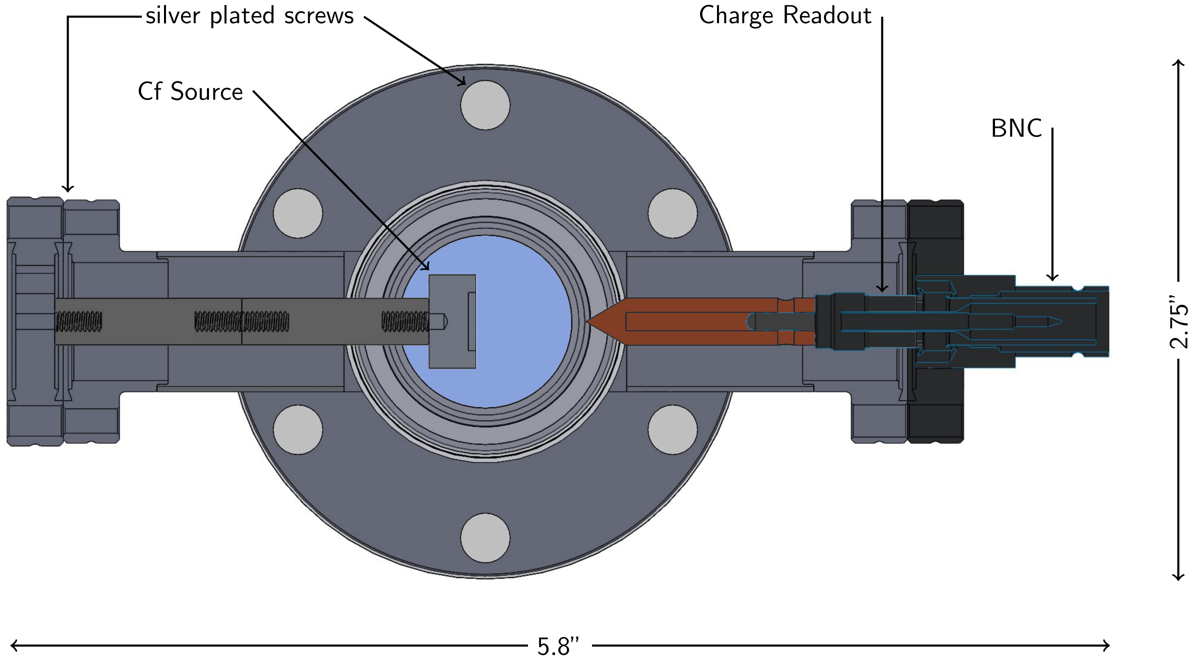
Secondly, before being filled with Xenon gas at around atmospheric pressure the lamp had to be evacuated bars with the aim of preventing contamination of the Xenon gas. This entailed several engineering choices.

Outgassing

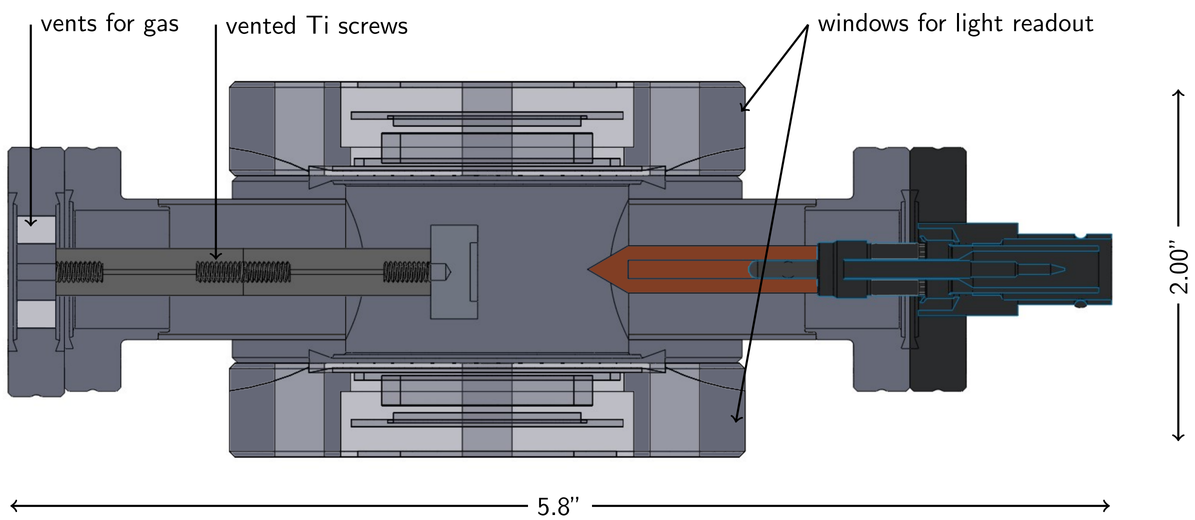
The effect of outgassing occurs when when gas that is trapped inside a material gets released and subsequently contaminates the vacuum or simply prevents further reduction of pressure. Outgassing rates depend on temperature, thus one can usually reduce the amount of contamination by "baking out" the vacuum system prior to evacuation.

Particular materials are especially well suited for vacuum application and where used in the current design as follows:

- 304L stainless steel: ConFlat®(CF) flanges and connecting pieces, casing of the Cf source



(a) Section view A: Vertical cut through the lamp (see figure 3.1a). The viewport has diameter of 2.75" with a 1.5" window in the centre, coloured light blue in the figure.



(b) Section view B: Horizontal cut through the lamp (see figure 3.1b). Gas is filled in from the left, while the charge is readout via the right sight. The light-port are on top and in the bottom of this rendering.

Figure 3.2.: Two section views of the final design of the lamp. All parts are to scale while the colours are chosen for better visibility.

- aluminum: standoff
- oxygen-free copper: gaskets for the CF flanges[2], extension of the pinch
- silver-coated, stainless steel screws for the connection of the flanges[3]
- titanium screws for the attachment of the Cf source

Outgassing can also happen from pockets of air enclosed during the assembly (e.g. in screw-holes etc). To prevent this from happening the entire lamp should form one single volume inside the lamp. In particular, one had to use a vented standoff to install the Cf source and vent the titanium screws by hand.

Cold welding

In the presence of air the surface of metals contain an oxide layer that is closed when broken. In the absence of oxygen this do not happen, thus two surfaces can weld together. This effect is also referred to as sticking, stiction, or adhesion. As a result, the parts can not be separated even if the the pressure is increased again, making a dis-assembly of the lamp impossible.

One therefore has to make sure that touching surfaces are made of materials that cannot weld together. In particular we made sure to use titanium or silver-plated screws.[17]

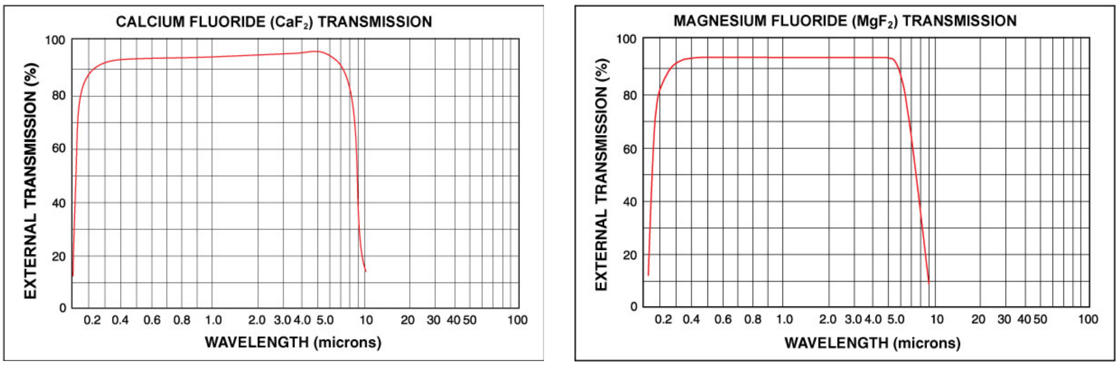
3.3.2. Absorption of Light

Light at 175-178 nm is far UV light whose observation is not particularly easy. Absorption of light before detection is mainly due to two obstacles: The windows of the lamp and water vapor.

Windows

Finding windows with acceptable transmission in the very UV range proved challenging. Eventually we settled for windows made of Magnesium-Fluoride (MgF_2) and Calcium-Fluoride (CaF_2). There respective transmission charts can be seen in figure 3.3.

[3]Not or only partially shown in figure 3.2.



(a) Calcium-Fluoride (b) Magnesium-Fluoride

Figure 3.3.: Transmission charts for the window produced by *Kurt J. Lesker Company* [24].

Water vapor

Water vapor absorbs far UV light, thus we had to make sure that the concentration of water vapor between the window and the PMT was low. We decided achieve this by creating a layer of Nitrogen gas in between. Instead of flushing the entire box with with Nitrogen I designed a ring that can be attached to the lamp and fits the PMT. This ring is constantly flushed with nitrogen, preventing the absorption of light in the gap between the two windows of the lamp and the PMT.

The final design of the lamp, including the two PMTs as well as the pinch-off tube for the evacuation and the Xenon inflow can be seen in figure 3.4.

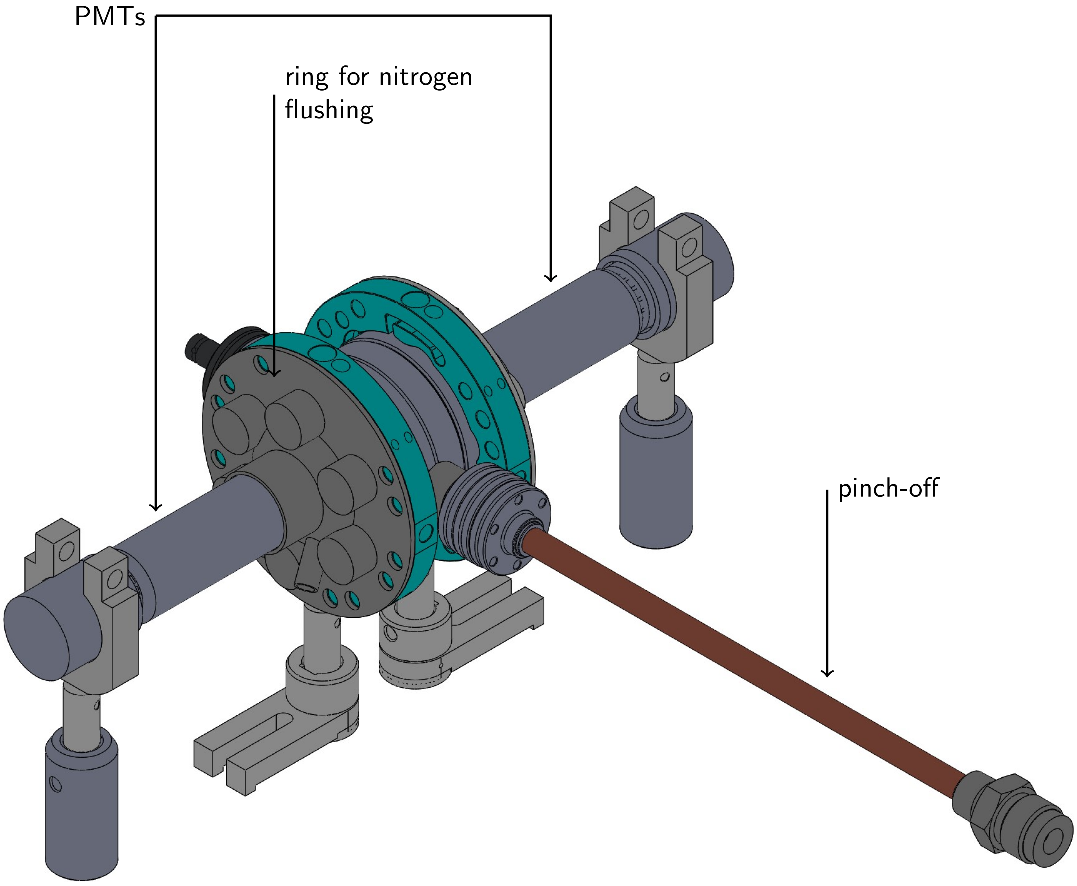


Figure 3.4.: 3D rendering of the setup including all relevant components.

3.4. Design study: Range of fission products in Xe gas

The production of scintillation light by particles travelling through matter depends on how and where the particle deposits its energy in the material. Therefore in the very beginning of the internship simulations on the energy deposit of the different fission products emerging from the decay of Cf-252 have been conducted using the software SRIM. The goal of those simulation was to determine the optimal position of the fission source within the scintillation chamber and a preliminary value for the pressure of the Xe gas inside the chamber.

The used parameters are summarized in 3.1 and will be explained in the following together with the results from the simulations.

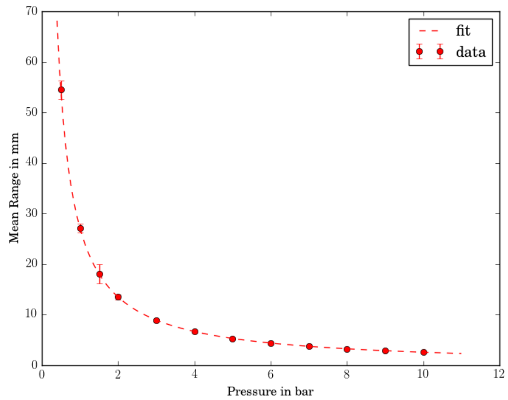


Figure 3.5.: Mean range of α-particles traveling through Xenon gas with hyperbolic fit superimposed.

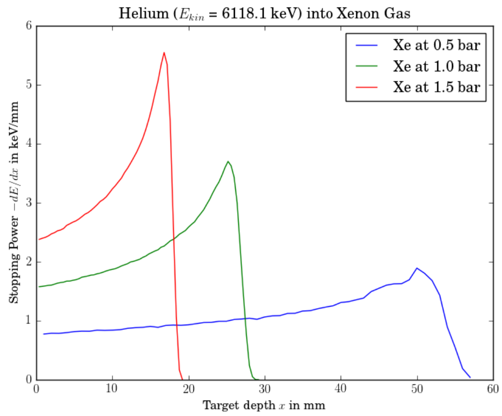


Figure 3.6.: Stopping power of α-particles in xenon gas at different pressures.

α-Decay

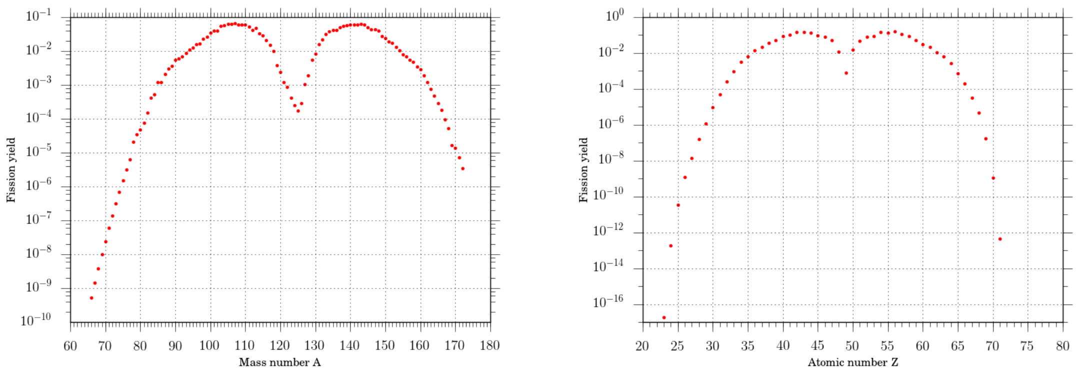
^{252}Cf primarily decays via the emission of α-particles to the unstable ground state of ^{248}Cm. The Q-value of the reaction is $Q_\alpha = 6216.87(4)\,\text{keV}$ [9], resulting in a kinetic energy of the α-particle of $E_\alpha = 6118.18\,\text{keV}$[4]

The stopping power for α-particles observes a Bragg peak just before the particles comes to rest. The position of the Bragg is highly dependent on the particle density inside the chamber, and thus the pressure of the gas. As the number of particles per volume increase so does the number of interactions per path length for the α-particles, decreasing distance they travel inside the chamber. Figure 3.5 shows the simulated mean distance of α-particles for pressures up to ten bars.

Spontaneous Fission

The spontaneous fission of ^{252}Cf happens via multiple decay channels. The spectrum of the fission yield can be seen in figure 3.7. The isotopes used for the simulations are Mo (index "L") and Xe (index "H"). They were found by associating the average masses of

[4]Using energy and momentum conservation one finds $E_\alpha = \frac{Q_\alpha}{1+m_\alpha/m_{\text{Cm}}}$, where m_α and m_{Cm} are the rest masses of the α-particle and the recoiling daughter nucleus.



(a) (b)

Figure 3.7.: Yield for the different products in the spontaneous fission of ^{252}Cf by their (a) mass number A and their (b) atomic number Z, respectively. Data obtained from [22]

the two runs presented in [16] with the closest fitting, complementary decay products[5] in the ^{252}Cf spectrum.

	Run 1	Run 2	Average
$\langle m_{\mathrm{L}} \rangle$ [a.m.u.]	106.16 ± 0.2	106.22 ± 0.2	106.18 ± 0.14
$\langle m_{\mathrm{H}} \rangle$ [a.m.u.]	142.17 ± 0.2	142.13 ± 0.2	142.15 ± 0.14
$\langle E_{\mathrm{L}} \rangle$ [MeV]	102.54 ± 0.94	102.62 ± 1.35	102.58 ± 0.82
$\langle E_{\mathrm{H}} \rangle$ [MeV]	78.68 ± 0.50	78.66 ± 0.71	78.67 ± 0.46

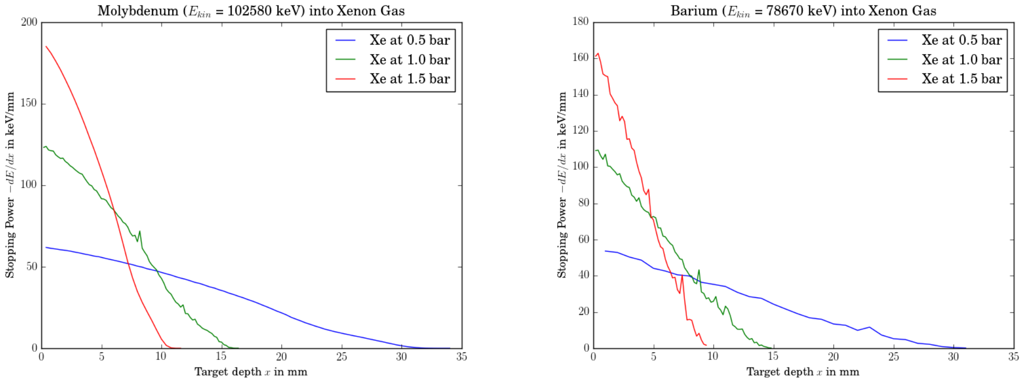
Table 3.1.: Caption

Unlike the α-particles, the distribution of $-\frac{dE}{dx}$ doesn't observe a Bragg peak for neither the heavy nor the light fission product. The particles loose most of their kinetic energy in close proximity of the source.

Conclusion for the Design

Analysis done by Ostrovskiy et al on behalf of the *nEXO* collaboration has shown that both α-decays and spontaneous fissions contribute significant numbers of photons (see figure 3.9).

[5] $A_{\mathrm{H}} + A_{\mathrm{L}} = 252$ and $Z_{\mathrm{H}} + Z_{\mathrm{L}} = 98$



(a) Light fission product (b) Heavy fission product

Figure 3.8.: Stopping power associated with the light (left) and heavy (fission) products at different pressures. The elements used were Mo and Ba, respectively.

For maximizing the light output of the lamp, one would want to use both decay modes and produce the majority of photons in close proximity centre of the chamber. Therefore, the decision was made to place the fission source slightly offset from the centre as shown in figure 3.2.

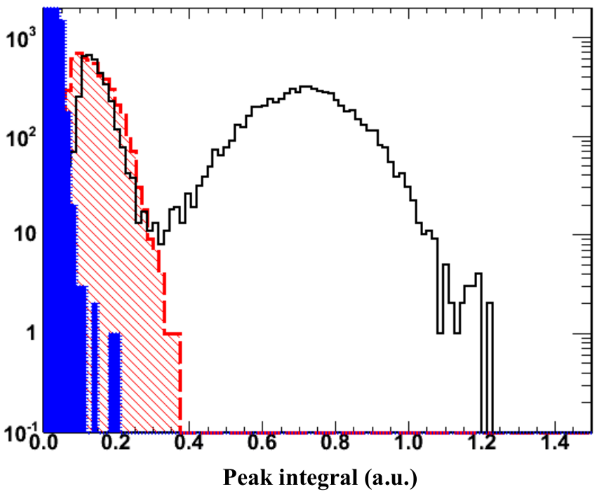


Figure 3.9.:
Charge spectrum recorded SiPM device (Model: FBK-2010), exposed to xenon scintillation light from two different fission light sources: ^{252}Cf (black line) and ^{241}Am (red hatched line). While ^{241}Am only undergoes α-decays, ^{252}Cf additionally observes spontaneous fissions. Those events produces a larger number of photons, resulting in a larger peak integral, visible as the second wider peak in the peak in the spectrum. The blue area shows the noise distribution of the device.[20]

4. Control of the digitizer

The data acquisition on the experiment will be done using a waveform digitizer that converts the analog readout signal from the different channels (PMTs, SiPMs or charge readout) into a digital signal that consequently can be processed into the data types described in chapter 2. The device used in the laboratory for the is a CAEN DT5730 8 Channel 14-bit 500 MS/s Desktop Waveform Digitizer. The control and communication of said digitizer was another major focus of my internship at McGill University.

4.1. Technical Specifications

The DT5730 has 8 channels with software select-able input range of 2 or 0.5 Volt (peak-to-peak). Using a 16-bit DAC on each channel the DC offset can be adjusted in the range $\pm 1\,\mathrm{V}$, allowing the user to record also unipolar input signals.

Each channel has a SRAM Multi-Event Buffer that can be divided $1 \sim 1024$ buffers. The device in use has a digital memory of 640 kS/ch, leading to a acquisition window between $1.8\,\mathrm{ms} \sim 1.8\,\mu\mathrm{s}$.

Tiggers can be provided externally through the TRG-IN input connector on the front panel or via a software trigger. Additionally the channels can generate self-trigger requests. The trigger from the board might be propagated outside through the GPO output connector. See section 4.3 for details.

Multi-board synchronization is supported by both the hardware/firmware of the board and the software framework designed during the internship.

Communication between the DT5730 and a computer can be done via USB 2.0 or an optical link (data transfer up to 30 MB/s and 80 MB/s, respectively). The data acquisition system was setup on a Linux device (Ubuntu 18.04) making use of the existing C libraries provided by the manufacturer.

4.2. Communication with the DT5730

Communication with the DT5730 is possible through several registers. Each register of the DT5730 has 32 bits, thus can store exactly one unsigned 32-bit integer (**`uint32_t`** or **`DWORD`**). Which function the different registers are assigned to depends on the firmware in use. For the presented work the "standard" firmware provided by *CAEN* as explained in [26] has been chosen. For an more detailed description please refer to this manual.

According to the definition in [26], registers like `0x1nXX` refers to an array of register where n can generally take values between 0 and 15 (0 and F in hex-decimal numbers). n often corresponds either channels or couple of channels[1]. If the concrete usage is not clear from the context in this documentation, please see [26]. If several couples of one array should have the same value, one can also reset register `0x8nXX`.

4.3. Trigger Management

The DT5730 has several available trigger sources that can be used to trigger data recording:

- External trigger provided through the GPO port on the front panel
- Software trigger produced by writing to register `0x8108`
- Self trigger signals produced by the channels

For board operated with the default software the acquisition trigger is a board common trigger, e.g a trigger generated on the motherboard that is then propagated to the individual channels to record the waveform. The following section will explain the logic behind the generation of self-trigger requests.

[1] two adjunct channels

4.3.1. Self-trigger logic

For the production of self trigger requests the eight channels are divided up into four distinct couples. Following the convention in [26] couple n corresponds to channels n and n+1 (thus only couples 0, 2, 4 and 6 are meaningful for the DT5730).

Each couples FPGA can be programmed such that the self-trigger for the two channels of the couple is either

- an over-threshold (under-threshold) signal that is active as long the signal is over (under) the set threshold in case of a positive (negative) trigger polarity, or
- a pulse signal with a defined length (set separately for both channels through register `0x1n70`).

From those signals, couple n may configured (through register `0x1n84` or `0x8084`) to produce a trigger requests in one of the following four cases:

(a) AND — both channel n and n+1 produce a signal

(b) ONLY N — only channel n produces a signal

(c) ONLY N+1 — only channel n+1 produces a signal

(d) OR — either channel n or channel n+1 produce a signal

Figure 4.1 illustrates the effect of those four different settings on a given input in the two adjunct channels 0 and 1.

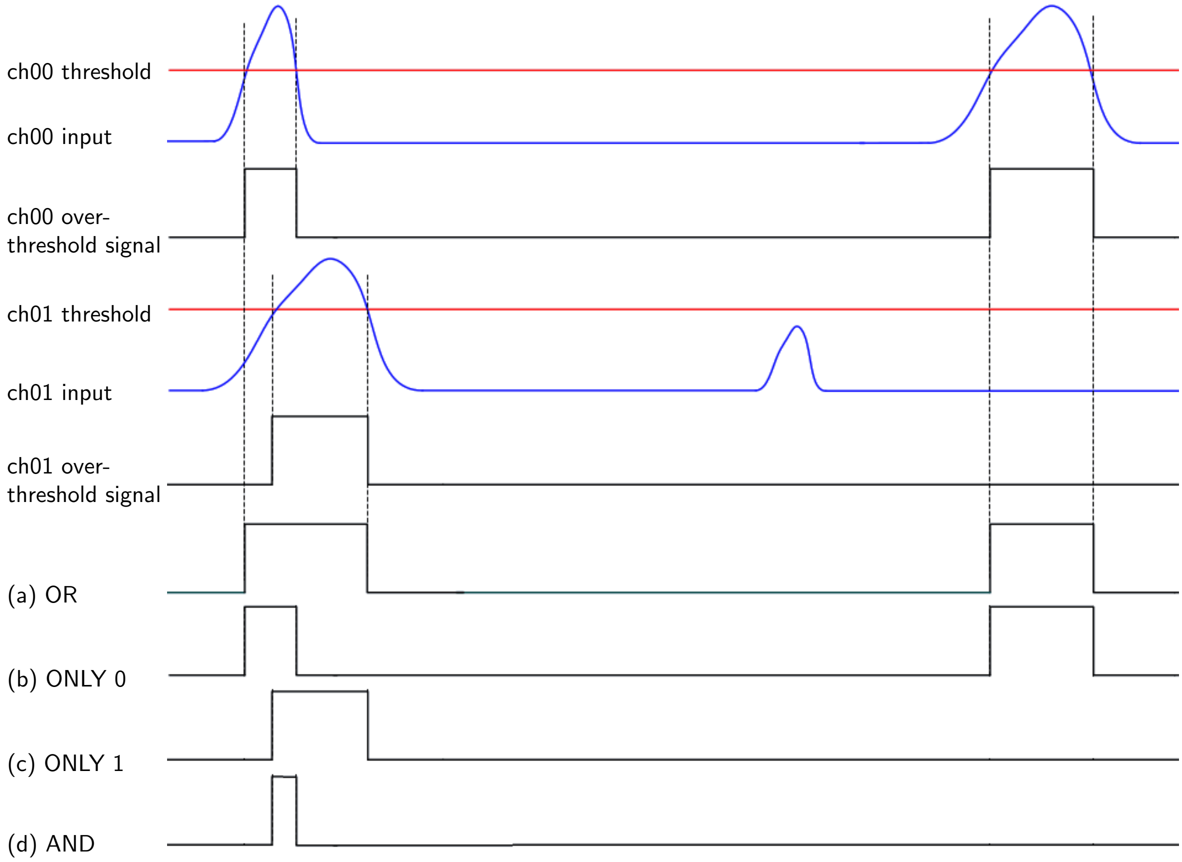


Figure 4.1.: Trigger signal of couple 0 for the four different trigger logic settings assuming an over-threshold signal (positive trigger polarity). From top to bottom: (a) OR, (b) ONLY 0, (c) ONLY 1 and (d) AND. Figure from [27] with modifications for this documentation.

4.3.2. Coincidence Measurements

Making coincidence measurements with the default software is possible through the majority operation. This requires modifications at register `0x0x810C`. One is able to modify

- the trigger requests taking part in the coincidence (bits [3:0], where bit n corresponds to couple 2n),
- the length of the coincidence window (T_{TVAW}) (bits[23:20] in steps of the ADC clock, e.g. 8 ns resulting in a length between 0 and 64 ns), and
- the majority level (m) (bits[26:24])[2].

A common trigger is generated at the end of T_{TVAW} if the number of enabled triggers within T_{TVAW} exceeds the majority level.

For $m = 0$ the coincidence acquisition mode is disabled and the value of T_{TVAW} is meaningless. The trigger will be generated as soon as one of the trigger request is enabled.

For $m \neq 0$ and $T_{TVAW} = 0$, $m + 1$ must be enabled in the exact same instant (with a resolution of 8 ns). Even if one trigger request gets enabled while a second trigger request is already enabled, no trigger will be produced on the motherboard.

Figure 4.2 shows how different combinations of m and T_{TVAW} affect the trigger generation. [20]

[2] The majority level needs to be smaller than the number of couples participating in the coincidence.

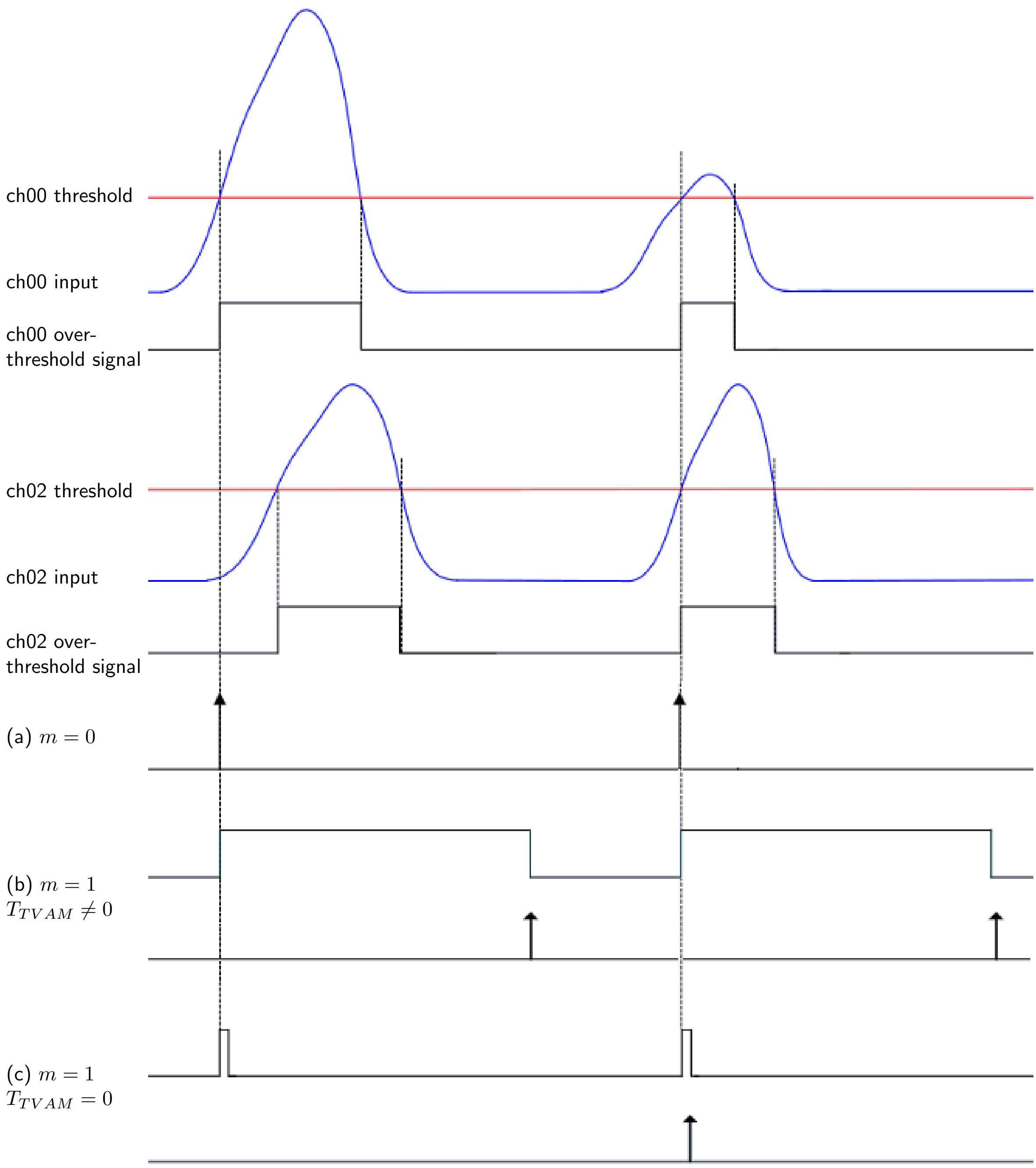


Figure 4.2.: Effect of three different different settings of majority level (m) and coincidence window $\mathrm{T_{TVAW}} \neq 0$. For simplifications it is assumed the only channel 0 and 2 are participating in the trigger generation (over-threshold signal, positive polarity). From top to bottom:
(a) $m = 0$, (b) $m = 1$ and $T_{TVAW} \neq 0$, and (c) $m = 1$ and $\mathrm{T_{TVAW}} = 0$

4.4. Analysis: Position measurement with using two PMTs

4.4.1. Goal and setup

The goal of this first simple experiment was to test the functionality of the digitizer, software, and the two purchased PMTs. The setup consisted of the two PMTs mounted onto a ruler facing each other at the two positions z_1 and z_2[3]. Between the two PMTs a blue LED was positioned (at position z_{LED}) on a movable post. The entire setup was leveled by hand such that the cylinder axis of the two PMTs coincided and the LED moved along this axis. The experiment was conducted inside a dark box to eliminate sources other than the LED.

4.4.2. Theory

The discharge of the individual PMT depends linearly on the number of incident photons at the respective PMT position (z_j). The proportionality constant is the detection efficiency of each PMT at the set voltage (ϵ_i). The voltage for optimal gain was determined beforehand and was kept constant over the experiment

For a completely isotropic light source, the number of incident photons is inversely proportional to the square of the distance between said PMT and the LED at position z_{LED}. We account for inhomogenities in the photon distribution by introducing the factor P_j that might arise from anisotropies of the light emission of the LED or reflections inside the box. Thus the integrated charge, e.g. the signal of PMT i at position z_j can be described as

$$Q_{ij}(z_{\text{LED}}) = \frac{\epsilon_i \, P_j}{(z_{\text{LED}} - z_i)^2} \times N_{\text{photons}} \tag{4.1}$$

where N_{photons} is the number of photons produced by the LED during one pulse. This number was assumed to be constant.

[3] $z_1 = 19.2$ cm and $z_2 = 70.8$ cm measured from the window of the PMT. Taking the middle between the two PMTs as reference this corresponds to ± 25.8 cm

From equation 4.1 one can easily calculate the "relative position" of the LED

$$\frac{z_{\text{LED}} - z_1}{z_{\text{LED}} - z_2} = \sqrt{\frac{\epsilon_j P_1}{\epsilon_i P_2} \times \frac{Q_{i2}(z_{\text{LED}})}{Q_{j1}(z_{\text{LED}})}} \equiv \beta \tag{4.2}$$

Solving 4.2 yields z_{LED} depending on the quantity to be measured, β

$$z_{\text{LED}} = \frac{z_1 - \beta z_2}{1 - \beta} \tag{4.3}$$

In this equation on still has to determine the ratio $\epsilon_j P_1/\epsilon_i P_2$ which has been done by two measurements where the LED was placed at z^0_{LED} in the middle between the two PMTs such that $|z^0_{\text{LED}} - z_1| = |z^0_{\text{LED}} - z_2|$. The PMTs where exchanged in between the measurements, enabling us to measure all four matrix elements

$$\mathbf{Q}(z^0_{\text{LED}}) = \begin{pmatrix} Q_{11}(z^0_{\text{LED}}) & Q_{12}(z^0_{\text{LED}}) \\ Q_{21}(z^0_{\text{LED}}) & Q_{22}(z^0_{\text{LED}}) \end{pmatrix} \equiv \begin{pmatrix} Q^0_{11} & Q^0_{12} \\ Q^0_{21} & Q^0_{22} \end{pmatrix} \propto \begin{pmatrix} \epsilon_1 P_1 & \epsilon_1 P_2 \\ \epsilon_2 P_1 & \epsilon_1 P_2 \end{pmatrix} \tag{4.4}$$

The ratio between the two efficiencies and position factors follows naturally as

$$\begin{aligned} \epsilon_1/\epsilon_2 =& Q^0_{11}/Q^0_{21} = & Q^0_{12}/Q^0_{22} \\ P_1/P_2 =& Q^0_{11}/Q^0_{12} = & Q^0_{21}/Q^0_{22} \end{aligned}$$

4.4.3. Measurement and Analysis

First we made two calibration runs in order to determine the ratio of the coefficients ϵ_i and P_j as described earlier by swapping the PMTs while the LED was placed in middle between the two PMTs. Figure 4.3 illustrates two waveforms superimposed in one window obtained during one of the calibration runs.

Subsequently the lamp was moved along the ruler between 25 cm and 65 cm ($\pm$20 cm, respectively).

The obtained sets of events where analyzed using a code provided from Dr. Thomas McElroy that was originally used by the *DEAP* collaboration and had been modified to fit into the software framework of the experiment. Summing over the ADC above baseline for each event one has a measure for the signal of the PMT and thus the number of photons incident to the PMT in the event.

This sum was used to

- determine the coefficients on an event-by-event basis from the two calibration runs
- determine the LED position according to equation 4.3 on an event-by-event basis
- illustrate the signal dependence on the (nominal) LED position (see figure 4.5)

All distributions were observed to be normally distributed and thus were fitted using a Gaussian fit with the mean corresponding to the value and the standard deviation corresponding to the error of the corresponding variable.

Figure 4.5 shows the depends of the PMT signal on the (nominal) LED position and illustrates that it follows the inverse square law as expected.

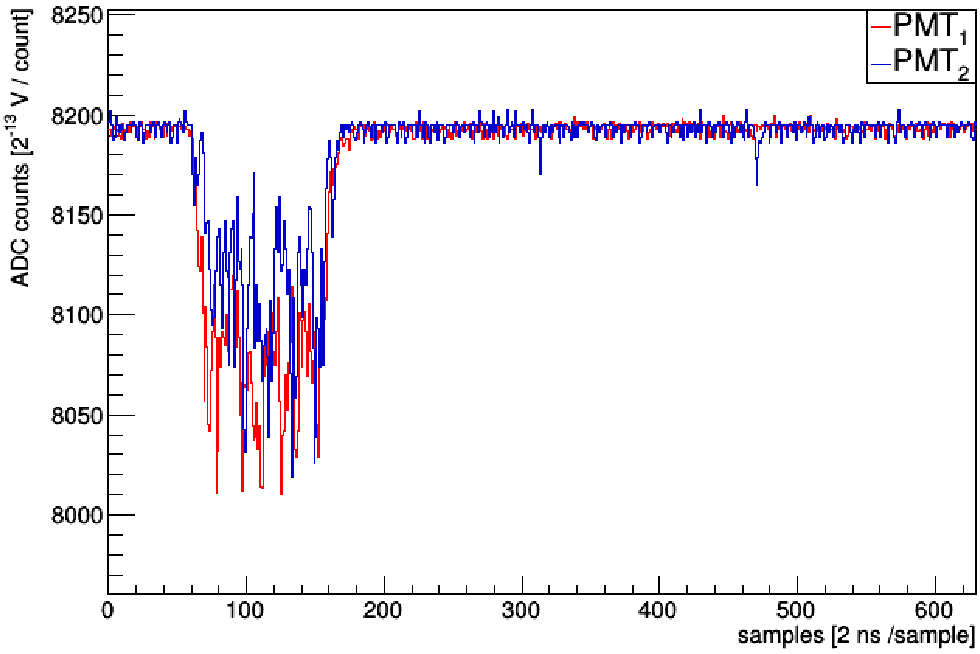


Figure 4.3.: Example waveform from one of the calibration runs showing the output of both PMTs for one event. Note the units of the axes. One can clearly recognize the time during which the LED was turned on

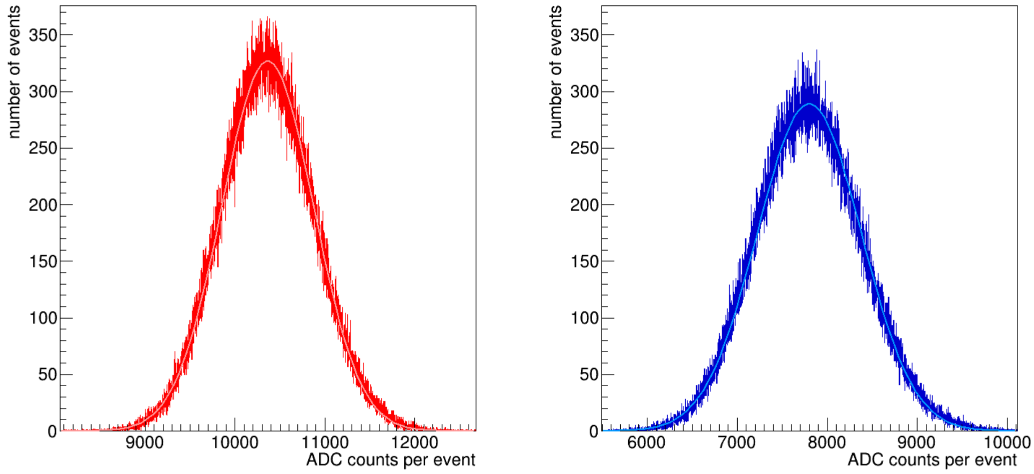


Figure 4.4.: Distribution of ADC counts per event for PMT_1 (red, left) and PMT_2 (blue, right) for one of the calibration runs. Superimposed is a Gaussian fit to the data.

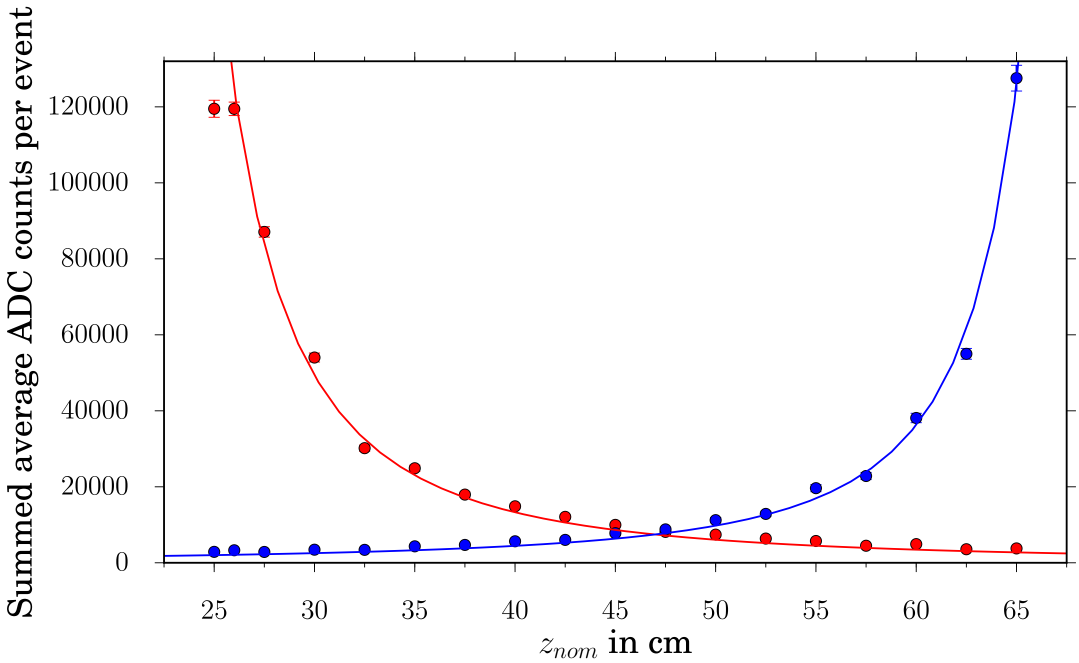


Figure 4.5.: Average of the summed ADC counts per events for PMT_1 in red and PMT_2 in blue. The average and its error were taken as the mean and standard deviation of a Gaussian fit to the measurements (see figure 4.4). One set consists of around 5×10^6 events. Superimposed is a hyperbolic fit.

4.4.4. Results and Discussion

Table 4.1 shows the results of the initial two calibration runs. In the following the average displayed in the last column was used. As mentioned before the value and error where found as the mean and standard deviation of a Gaussian fit to the data obtained on an event-by-event basis.

	Q^0_{11}/Q^0_{ij}	Q^0_{ij}/Q^0_{22}	average
ϵ_1/ϵ_2	1.3288 ± 0.1091	1.3035 ± 0.1072	1.3162 ± 0.0765
P_1/P_2	0.9927 ± 0.0691	0.9738 ± 0.1007	0.9833 ± 0.0611

Table 4.1.: Ratios of the coefficients

Figure 4.6 shows the results of the position measurements. Despite the agreement of the measurements with the inverse square law as shown in figure 4.5 we were not able to reproduce the nominal LED position the further away from the middle between the two PMTs we moved. Due to time constraints no further investigation of this difference between experiment and theory was undertaken

Nevertheless, the experiment can be regarded as successful. The main goal of demonstrating and testing the functionality interplay of soft- , firm- and hardware was achieved.

Notably, the use of coincidence measurements (see section 4.3.2) made it possible to push the signal threshold down to single photon level ($\sim$ 15 ADC below/above baseline) without picking up significant noise. Measurements at the very extreme positions where one of the PMTs did receive very little light were thus possible (see figure 4.7).

Several issues and bugs concerning the setup of the digitizer were discovered and solved during the these test runs. One example was the realization that the baseline of the different channels were not only not equal but also shifting between different runs making a reliable setting of trigger thresholds particularly hard. This hardware problem was tackled by software fix that allows the user to check and optionally adjust the baseline prior to starting a run.

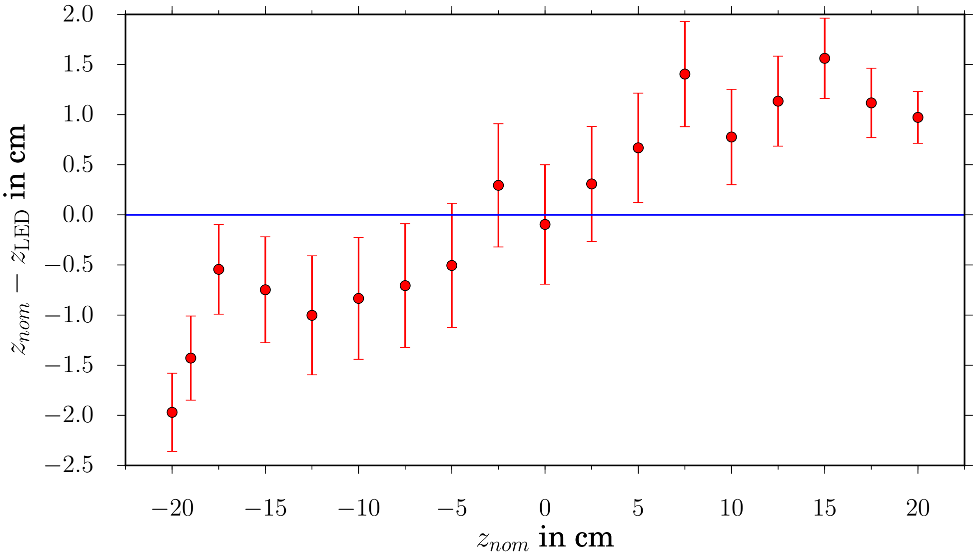


Figure 4.6.: Deviation of the measured (z_{LED}) from the nominal (z_{nom}) LED position from a Gaussian fit to the distributions.

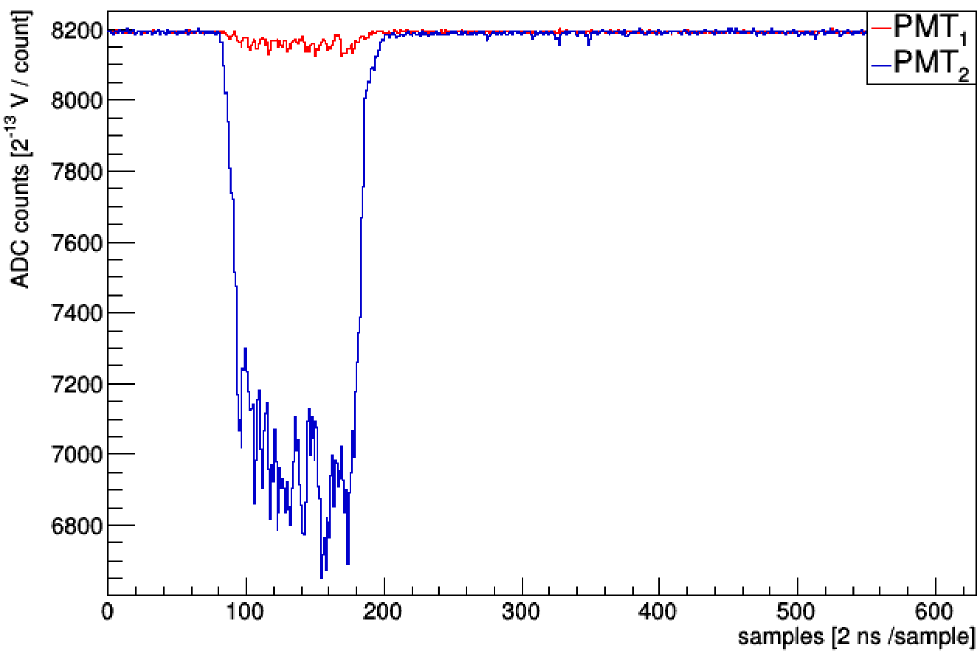


Figure 4.7.: Waveform for one event where the LED was placed very close to PMT_2, at position $z_{\mathrm{LED}} = 65\,\mathrm{cm}$.

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I want to express my gratitude to my supervisors Prof. Dr. Thomas Brunner and Dr. Thomas McElroy for the opportunity to work at *McGill University*, for their engagement, their support and their trust during my four months in Montreal.

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A. Data Classes

class: **RawBlock**	
protected attributes	
uint32_t	*startBin*
std::vector<uint16_t>	*samples*
public member functions	
	RawBlock()
	RawBlock(**const RawBlock&** *RawBlock*)
	RawBlock()
RawBlock&	operator=(**const RawBlock&** *RawBlock*)
void	SetSamples(**std::vector<uint16_t> &_samples**)
Int_t	GetSampleCount()
bool	ExistSamples()
void	PruneSamples()
void	SetStartBin(**uint32_t** *_startBin*)
uint32_t	GetStartBin()
protected member functions	
void	Init()
void	Destroy()
void	CopyObj(**const RawBlock&** *RawBlock*)

Table A.1.: RawBlock

class: **Pulse**	
protected attributes	
Long_t	*chargeADC*
uint16_t	*minimumADC*
uint16_t	*left*
uint16_t	*right*
uint16_t	*referenceADC*

std::vector<uint16_t>	*peakvec*
std::vector<uint16_t>	*phase*
Long_t	*baselineIntRel3892*
Long_t	*baselineSquareIntRel3892*
Long_t	*baselineSamples*
Byte_t	*boardID*
Byte_t	*inputID*
uint16_t	*blockStartBin*
std::vector<Float_t>	*fitFactor*
std::vector<Float_t>	*width20_to_20*
uint16_t	*peakCharge*
Float_t	*maxD*
uint16_t	*maxV*
uint16_t	*maxVT*
	public member functions
	Pulse()
	virtual ~Pulse() **const**
virtual TObject*	Clone(**const char*** *opt* = "") **const**
uint16_t	GetWidth()
void	SetWidth20to20(**std::vector<Float_t>** *_width20_to_20*)
Float_t	GetWidth20to20(**uint16_t** *ipeak*)
Int_t	GetPeak(**int** *n*)
Double_t	GetBaselineRMS()
Double_t	GetBaseline()
Long_t	GetBaselineSamples()
uint16_t	GetSubpeakCount()
Float_t	GetSubpeakTime(**UInt_t** *n*)
Float_t	GetCharge()
ULong_t	GetChargeRaw()
void	SetCharge(**ULong_t** *_chargeADC*)
void	SetMinimunADC(**uint16_t** *_minimum*)
uint16_t	GetMinimumADC()

Float_t	GetPeakHeightADC()
void	SetLeftEdge(**uint16_t** *_left*)
int	GetLeftEdge()
void	SetRightEdge(**uint16_t** *_right*)
int	GetRightEdge()
void	AddSubpeakMinimum(**uint16_t** *_minimum*)
uint16_t	GetSubpeakMinimum()
void	PruneSubpeakMinima()
void	AddSubpeak(**uint16_t** *_peak*)
void	SetSubpeaks(**std::vector<uint16_t>** *_peaks*)
int	GetSubpeak(**UInt_t** *n*)
Short_t	GetPhase()
Short_t	GetPhase(**UInt_t** *n*)
void	AddPhase(**Short_t** *_phase*)
void	SetPhases(**std::vector<uint16_t>** *_phase*)
Float_t	GetFitFactor(**UInt_t** *n*)
void	SetFitFactors(**std::vector<Float_t>** *_factor*)
void	SetFitFactor(**UInt_t** *n*, **Float_t** *_factor*)
void	AddFitFactor(**Float_t** *_factor*)
void	PruneFitFactor()
void	AddChi2(**std::vector<Float_t>** *_Chi2*)
Float_t	GetChi2(**UInt_t** *subpeaknumber*, **int** *phaseIndes*)
void	SetTimingSamples(**std::vector<uint16_t>** *_SubnsSamples*)
void	ResetTimingSamples()
Float_t	GetTimingSample(**UInt_t** *subpeakNumber*, **int**, numSamplesFromPeak)
void	SetSubpeakMinima()
Float_t	GetTotalFitCharge()
Float_t	GetAdjustedFitCharge()
Float_t	GetSubpeakCharge(**UInt_ti** *)*
Float_t	GetTotalSubpeakCharge()

Float_t	GetAdjustedFitChargeADC(**UInt_ti**)
Float_t	GetSubpeakChargeADC(**UInt_ti**)
Float	GetFitChargeADC(**UInt_ti**)
void	SetBaselineIntRel3892(**Short_t** _baselineIntRel3892)
void	SetBaselineSquareIntRel3892(**Int_t** _baselineSquareIntRel3892)
void	SetBaselineSamples(**Long_t** _baselineSamples)
Long64_t	GetBaselineSumOfSquares()
void	SetRawBlockStartBin(**uint16_t** _blockStartBin)
uint16_t	GetRawBlockStartBin()
void	PruneSubpeakUnderCharge(**Float_t** cut)
void	SetMaxVT(**UShort_t** _maxVT)
void	SetPeakChargeADCRel4096(**UShort_t** _peakchargeADC)
void	SetMaxD(**Float_t** _maxD)
void	SetMaxV(**uint16_t** _maxV)
void	SetReferenceADC(**uint16_t** _referenceADC)
uint16_t	GetReferenceADC()
protected member functions	
void	Init()
void	Destroy()

Table A.4.: Pulse

class: **Channel**	
protected attributes	
uint32_t	*temp*
int	*ID*
std::vector <RawBlock>	*waves*
public member function	
	Channel()
	Channel(**const Channel&** *ch*)
	∼Channel()
Channel&	operator=(**const Channel&** *ch*)
int&	GetChID()
void	SetChID(**int&** *_ID*)
uint32_t	GetTemp()
void	SetTemp(**uint32_t&** *_temp*)
RawBlock	GetWave(**const int&** *numWave*)
RawBlock*	GetWavePoint(**const int&** *numWave*)
void	SetWaves(**std::vector<RawBlock>&** *_waves*)
void	AddWave(**RawBlock&** *wave*)
Int_t	GetWaveCount()
bool	ExistWaves()
void	PruneWaves()
protected member functions	
void	Init()
void	Destroy()
void	CopyObj(**const Channel&** *ch*)

Table A.2.: Channel

class: **TriggerInfo**	
protected attributes	
uint64_t	*trg_time* time within the current run in units of the ADC clock (i.e. 8 ns)
public member functions	
uint64_t&	GetTriggerInfo()
void	SetTriggerInfo(**uint64_t&** *_trg_time*)
protected member functions	
void	Init()
void	Destroy()
void	CopyObj(**const TriggerInfo&** *trg*)

Table A.3.: TriggerInfo

class: **PMT**	
protected attributes	
Short_t	*id*
Float_t	*qTotal*
Float_t	*qPE*
std::vector<Pulse*>	*pulse*
Float_t	*tofOffset*
public member functions	
	PMT()
	PMT(**const PMT&** *rhs*)
	~PMT()
PMT&	operator=(**const PMT&** *rhs*)
virtual TObject*	Clone(**const char*** *opt* = "") **const**
void	SetID(**Int_t** *idum*)
Int_t	GetID()
void	SetTotalQ(**Double_t** *qTotalum*)
Double_t	GetTotalQ()
void	SetQPE(**Double_t** *qPEum*)
Double_t	GetQPE()
void	SetTOFOffset(**float** *tofOffsetum*)
Float_t	GetTOFOffset()
Int_t	GetPulseCount() **const**
Pulse*	GetPulse(**Int_t** *i*) **const**
void	ReplacePulse(**Int_t** *i*, **Pulse*** *p*)
Pulse*	GetLastPulse() **const**
Pulse*	AddNewPulse()
void	AddPulse(**Pulse*** *p*)
void	RemovePulse(**int** *ipulse*)
void	PrunePulse()
protected member functions	
void	Init()
void	Destroy()
void	CopyObj(**const PMT&** *rhs*)

Table A.5.: PMT

class: **Event**	
protected attributes	
TriggerInfo	*info*
int	*evtID*
std::vector<Channel>	*channels*
std::vector<PMT>	*pmts*
TVector3	*position*
public member functions	
	Event()
	Event(**const Event&** *evt*)
	~Event()
Event&	operator=(**const Event&** *evt*)
int&	GetEvtNum()
void	SetEvtNum(**int&** *_evtID*)
TriggerInfo&	GetTriggerInfo()
void	SetTriggerInfo(**TriggerInfo&** *_info*)
std::vector<Channel>&	GetChannels()
Channel*	GetChannel(**int** *i*)
void	SetChannels(**std::vector<Channel>&** *_channels*)
Int_t	GetChannelCount()
bool	ExistChannels()
void	SetPMTs(**std::vector<PMT>&** *_pmts*)
void	AddPMT(**PMT*** *pmt*)
PMT*	GetPMT(**int** *i*)
Int_t	GetPMTCount()
void	PrunePMTs()
protected member functions	
void	Init()
void	Destroy()
void	CopyObj(**const Event&** *evt*)

Table A.6.: Event